

Prime Fibernet

Enterprise Mobile Platform

Product Requirements Document

Version	2.0 — Production Ready
Date	June 2026
Platform	React Native + Expo Go
Audience	Customers · Officers · Admins (Unified App)
Status	Production Ready

Customer App

Officer App

Admin Panel

Executive Summary

Prime Fibernet Enterprise is a next-generation unified mobile and web application built with **React Native + Expo Go**, designed to manage broadband services with enterprise-grade security, zero-bug architecture, and powerful control mechanisms. This PRD defines the complete feature set for three integrated user roles operating within a single codebase.

Key Advantage	Description
Single Codebase	Supports iOS, Android, and Web (via Expo Web) from one repository
Faster Cycles	Hot reload enables rapid iteration and shorter release cycles
Native Performance	Zero compromise on speed or user experience
OTA Updates	Over-the-Air updates bypass app store approval delays
Enterprise Security	Multi-layer security at every application tier
Zero Downtime	Production-tested architecture with continuous availability

Product Overview

Prime Fibernet Enterprise consolidates three user personas into one elegant platform, each with distinct responsibilities and goals.

Customer App	Role: End Users	iOS · Android · Web
✓ Browse and compare broadband plans		
✓ Subscribe to plans and manage subscriptions		
✓ Track service requests in real-time		
✓ Make payments securely		
✓ Access customer support via AI chatbot		

Officer App	Role: Field Technicians	iOS · Android
✓ View assigned service requests with GPS navigation		
✓ Update request status and add activities		
✓ Manage shift attendance with location-based check-in		
✓ Track inventory assignments		
✓ Communicate with customers		
✓ View payslips and performance bonuses		

Admin Panel		Role: Administrators & Operations Managers	Web + Mobile (Responsive)
✓	Manage users — CRUD, block/unblock, billing		
✓	Manage officers — onboarding, assignments, payroll		
✓	Control subscription plans		
✓	Send bulk notifications		
✓	Generate comprehensive analytics and reports		
✓	Monitor system health and security		

Feature Specifications

PHASE 1: CORE (MVP — Weeks 1-12)	
Authentication & Security	
	✓ Email/password signup and login
	✓ JWT token-based authentication
	✓ 2FA for admin accounts (TOTP-based)
	✓ Password reset with email verification
	✓ Role-based access control (RBAC)
	✓ Session management with auto-logout
	✓ Biometric auth (fingerprint/face) for mobile
User Management (Admin)	
	✓ View all users with advanced filters
	✓ Block/unblock users with reason tracking
	✓ Edit user profile information
	✓ View subscription history and renewal dates
	✓ Send notifications to users
	✓ Export user data (CSV/PDF)
	✓ User activity audit trail
Officer Management (Admin)	
	✓ Invite officers via email
	✓ Officer onboarding workflow
	✓ Assign officers to regions
	✓ View officer availability status
	✓ Block/unblock officers
	✓ Manage officer salary configuration
	✓ View officer performance metrics
Plans Management	
	✓ Create, read, update, delete plans
	✓ Set plan pricing and validity
	✓ Define plan features
	✓ Activate/deactivate plans
	✓ Bulk import plans
	✓ Real-time plan sync across apps
Subscription Management (Customer)	
	✓ Browse all available plans
	✓ Filter plans by speed, price, category

	✓ View plan details and features
	✓ Subscribe to new plan
	✓ View active subscriptions
	✓ Renew subscription before expiry
	✓ Auto-renewal toggle
Payment Processing	
	✓ Multiple payment gateways (UPI, Cards, Net Banking, Wallets, Razorpay, Easebuzz)
	✓ Secure payment checkout flow
	✓ Payment history and receipts
	✓ Refund management
	✓ Invoice generation (PDF and email)
	✓ Payment status tracking
	✓ Automated billing reminders
Service Requests	
	✓ Create service requests (Installation, Repair, Upgrade, Complaint)
	✓ Assign requests to officers automatically or manually
	✓ Track request status in real-time
	✓ Add activities/notes to requests
	✓ Upload attachments (photos, documents)
	✓ SLA monitoring and alerts
	✓ Customer notifications on status updates

PHASE 2: FIELD OPERATIONS

Officer Dashboard

	✓ Daily metrics — assigned, completed, pending requests
	✓ Real-time request notifications
	✓ Quick stats: response time, completion rate
	✓ Shift status and attendance
	✓ Performance bonus tracker
	✓ Earnings overview

Request Management (Officer)

	✓ View assigned requests in list/map view
	✓ Filter by status, priority, date range
	✓ Open request details with full history
	✓ Update status: Pending → Assigned → In Progress → Completed
	✓ Add notes and attachments
	✓ Capture photos before/after

	✓ GPS-enabled location tracking
Location-Based Features	
	✓ Map integration (Google Maps / MapBox)
	✓ Navigate to customer location
	✓ Real-time location sharing with admin
	✓ Geofence-based check-in/out
	✓ Distance and time tracking
	✓ Location history reports
Shift Management	
	✓ View assigned shifts
	✓ Clock in/out with location verification
	✓ Location-based auto check-in at site
	✓ Track working hours
	✓ Overtime calculation
	✓ Leave request management
	✓ Shift approval workflow
Inventory Management	
	✓ View assigned inventory items
	✓ Track equipment: routers, modems, cables
	✓ Item condition tracking (Good, Damaged, Lost)
	✓ Quantity management
	✓ Serial number tracking
	✓ Return management
	✓ Inventory transfer between officers

PHASE 3: ADVANCED FEATURES**Analytics & Reporting**

	✓ Dashboard with KPIs
	✓ Customer acquisition trends
	✓ Revenue analytics by plan
	✓ Officer performance metrics
	✓ Request completion rates
	✓ SLA compliance reports
	✓ Export reports (PDF, Excel, CSV)
	✓ Scheduled report delivery via email
Notifications	
	✓ Push notifications (Firebase FCM)

	✓ Email notifications
	✓ SMS notifications
	✓ In-app notifications
	✓ Notification history
	✓ Notification preferences per user
	✓ Bulk notification sending
	✓ Notification templates
AI Chatbot Support	
	✓ RAG-based chatbot using Google Gemini
	✓ Knowledge base integration
	✓ Natural language understanding
	✓ Quick replies and suggestions
	✓ Escalation to human support
	✓ Chatbot conversation history
	✓ Learning and improvement logs
Payroll & HR	
	✓ Officer payslip generation
	✓ Salary configuration per officer
	✓ Overtime payment calculation
	✓ Performance bonus system
	✓ Leave tracking with LOP deduction
	✓ Bank account details management
	✓ Payslip download (PDF)
	✓ Salary advance requests
Invoice System	
	✓ Automated invoice generation
	✓ PDF invoice download
	✓ Email invoice delivery
	✓ WhatsApp invoice delivery
	✓ Invoice history
	✓ Payment tracking per invoice
	✓ GST/Tax configuration
	✓ Invoice customization (company logo, details)
Communication	
	✓ Officer-Customer messaging
	✓ WhatsApp integration for notifications
	✓ Email templates

	✓ SMS gateway integration
	✓ Bulk message campaigns
	✓ Message templates library
	✓ Message delivery tracking
PHASE 4: ENTERPRISE FEATU	
Advanced Security	
	✓ Data encryption (AES-256)
	✓ API rate limiting
	✓ DDoS protection
	✓ SQL injection prevention
	✓ XSS protection
	✓ CSRF tokens
	✓ Security audit logs
	✓ Compliance (GDPR, privacy standards)
Multi-Tenancy (Future Roadmap)	
	■ Support for multiple companies
	■ Isolated data per tenant
	■ Custom branding per tenant
	■ Tenant-specific configurations
	■ Consolidated analytics
Mobile-Specific Features	
	✓ Offline mode with sync
	✓ Low-bandwidth optimization
	✓ Background task processing
	✓ Push notification handling
	✓ App crash reporting
	✓ Performance monitoring
	✓ Battery optimization
Admin Settings	
	✓ Company information management
	✓ Email configuration (SMTP, domain)
	✓ Payment gateway settings
	✓ SMS gateway credentials
	✓ General settings (timezone, currency)
	✓ Feature flags and toggles
	✓ API keys management

Security Requirements

Domain	Requirement	Detail
Authentication	JWT Expiry	30-minute token expiry with refresh rotation
Authentication	Token Storage	Secure Keychain/Keystore on device
Authentication	Multi-Device Logout	Logout from all active sessions
Authorization	RBAC	Customer / Officer / Admin role separation
Authorization	Resource Access	Permission-based with row-level DB security
Authorization	Audit Logging	All critical actions logged with actor info
Data Protection	Encryption in Transit	TLS 1.3 for all API communication
Data Protection	Encryption at Rest	AES-256 + PII fields encrypted separately
Data Protection	Password Hashing	bcrypt with 10+ rounds
Data Protection	GDPR Compliance	Data retention, deletion, and export policies
Compliance	Security Audits	Regular penetration testing and vulnerability scans
Compliance	Data Residency	Region-specific data storage requirements met

Key User Stories

Customer Subscribes to Plan

As a customer, I want to browse plans and subscribe so that I can get broadband service.

Acceptance Criteria	Requirement
Authentication	User must be logged in
Plan List	All active plans displayed with price, speed, and validity
Plan Filter	Filterable by speed, price range, and category
Payment	Secure checkout with multiple gateway options
Confirmation	Email/SMS confirmation on successful subscription
Dashboard Update	Active subscription reflected immediately in dashboard

Officer Completes Service Request

As an officer, I want to view and complete assigned requests so that I can serve customers efficiently.

Acceptance Criteria	Requirement
Assignment	Officer receives push notification for new assignments
Map View	Request location shown on map with navigation option
Status Update	Officer can update status at each stage
Photo Capture	Before/after photos can be uploaded
Customer Notification	Customer notified at each status change
Completion	Request marked complete with timestamp and officer signature

Admin Generates Analytics Report

As an admin, I want to generate and export reports so that I can monitor business performance.

Acceptance Criteria	Requirement
Date Range	Configurable reporting period
KPI Dashboard	Revenue, subscriptions, officer performance at a glance
Export Formats	PDF, Excel, and CSV download options
Scheduled Reports	Automated delivery to email on set schedule
Drill-Down	Click-through from summary to detailed records

Development Timeline

Phase	Duration	Focus Areas	Deliverable
Phase 1 — Core MVP	Weeks 1–4	Auth, Users, Plans, Subscriptions, Payments, Service Requests	Functional MVP across all 3 roles
Phase 2 — Field Ops	Weeks 5–8	Officer dashboard, GPS, shift management, inventory	Full officer workflow live
Phase 3 — Advanced	Weeks 9–12	Analytics, AI chatbot, payroll, invoicing, bulk comms	Enterprise feature parity
Phase 4 — Enterprise	Weeks 13+	Security hardening, mobile optimisation, admin settings, Production ready release	Production ready release